

## 1. Goals of Security

**Ans:**

The main goals are **Confidentiality, Integrity, and Availability**.

This means data should be **kept secret**, remain **unchanged and accurate**, and be **accessible** to authorized users whenever needed.

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## 2. Authentication and Authorization

**Ans:**

**Authentication** checks *who the user is* — for example, entering a username and password.

**Authorization** checks *what that user can do* — like admin can edit, while normal users can only view data.

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## 3. Non-Repudiation

**Ans:**

It ensures that a sender **cannot deny sending** a message and receiver **cannot deny receiving** it.

This is mainly achieved through **digital signatures**, which prove the origin of the message.

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## 4. Vulnerability, Threat, and Attack

**Ans:**

A **vulnerability** is a weakness in the system,

a **threat** is something that might exploit it,

and an **attack** is when someone actually uses the weakness to harm the system or steal data.

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## 5. Classification of Cryptography

**Ans:**

1. **Symmetric key** – same key for encryption and decryption.
2. **Asymmetric key** – uses two keys: public and private.
3. **Hash function** – no key used; only produces a fixed-size output (hash).



## 6. Substitution and Transposition Cipher

**Ans:**

**Substitution** replaces each letter with another letter (Example: Caesar cipher – A→D).

**Transposition** only changes the order of letters (Example: HELLO → LLOHE).

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## 7. Monoalphabetic and Polyalphabetic Cipher

**Ans:**

In **monoalphabetic**, one alphabet set is used for all letters (Example: Caesar cipher).

In **polyalphabetic**, multiple alphabet sets are used, making it harder to crack (Example: Vigenère cipher).

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## 8. Stream and Block Cipher

**Ans:**

**Stream ciphers** encrypt data one bit or byte at a time (Example: RC4).

**Block ciphers** encrypt data in fixed-size blocks, like 64 or 128 bits (Examples: DES, AES, RC5).

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## 9. Product Cipher

**Ans:**

A product cipher combines **substitution** and **transposition** to make encryption stronger.  
Example – DES is a product cipher because it uses both techniques together.

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## 10. Symmetric vs Public Key Cryptography

**Ans:**

**Symmetric** uses the same key for both encryption and decryption — it's fast but less secure.

**Public key (Asymmetric)** uses two keys: one public, one private — more secure but slower.

Examples: DES (symmetric), RSA (public key).

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## 11. DES (Data Encryption Standard)

**Ans:**

DES encrypts data in **64-bit blocks** using a **56-bit key** with **16 rounds** of processing. It is **fast** but considered **weak** today because the key size is small and can be brute-forced.

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## 12. AES (Advanced Encryption Standard)

**Ans:**

AES is the improved version of DES. It uses **128, 192, or 256-bit keys** and works in multiple rounds.

It is **fast, strong, and widely used** in modern encryption systems like WhatsApp or VPNs.

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## 13. Session Key

**Ans:**

A **session key** is a temporary key used for one communication session only. It ensures that even if the key is stolen, only that single session is affected.

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## 14. RSA Algorithm

**Ans:**

1. Choose two prime numbers ( $p, q$ ) and find  $n = p \times q$ .
  2. Choose  $e$  and calculate  $d$  using a formula.
  3. Public key =  $(e, n)$ , Private key =  $(d, n)$ .  
Used for **encryption and digital signatures**.  
It's **very secure** but **slower** due to large keys.
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## 15. Digital Signature

**Ans:**

It ensures **authentication and message integrity**.

The sender creates a **hash** of the message and encrypts it with their **private key**.

Receiver decrypts it using the sender's **public key** — if hash matches, message is original.

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## 16. Hash Functions

**Ans:**

Hash functions take input and produce a **unique fixed-size output** called a hash value.  
Used for **passwords, integrity checking, and digital signatures**.  
Even a small change in input completely changes the hash.

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## 17. Digital Certificate

**Ans:**

Issued by a **Certificate Authority (CA)** to prove identity.  
Contains details like name, public key, validity, and CA's signature.  
Used in **HTTPS websites** to build trust and ensure authenticity.

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## 18. Kerberos

**Ans:**

Kerberos is a **network authentication protocol** that uses a **ticket system**.  
It verifies users securely without sending passwords over the network.  
Commonly used in **corporate or Windows networks**.

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## 19. IPSec

**Ans:**

IPSec secures data at the **network layer** by **encrypting IP packets**.  
It ensures **confidentiality and integrity** during data transfer.  
Commonly used in **VPNs**.

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## 20. Firewall

**Ans:**

A firewall acts as a **barrier between trusted and untrusted networks**.  
It filters traffic and blocks unauthorized access.  
Types – Packet filter, Proxy, and Stateful firewall.

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## 21. Intrusion Detection System (IDS)

**Ans:**

IDS **monitors network or system** activities for suspicious behavior or attacks.

Types – **Host-based IDS (HIDS)** and **Network-based IDS (NIDS)**.

It alerts administrators when unusual activity is detected.

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## 22. Diffie-Hellman Key Exchange

**Ans:**

It's used to **securely exchange secret keys** between two parties over an insecure channel.

It does not share the key directly but helps generate the same key at both ends.

Used in **SSL, HTTPS, and VPN**.

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## 23. Malware Types

**Ans:**

- **Virus:** Attaches to files and spreads.
  - **Worm:** Spreads automatically through networks.
  - **Trojan:** Pretends to be safe software.
  - **Rootkit:** Hides other malware.
  - **Keylogger:** Records what you type.
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## 24. DoS Attack

**Ans:**

A **Denial of Service (DoS)** attack floods a server with too many requests.

This makes the server crash or become very slow.

Example: **Ping of Death** or **HTTP Flood**.

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## 25. Brute Force Attack

**Ans:**

In this attack, the hacker tries **all possible keys or passwords** until one works.

It's **simple but time-consuming**, especially with long passwords.

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## 26. Cryptanalysis

**Ans:**

It means **breaking or decoding encrypted data** without knowing the key.

Example – Using **frequency analysis** to break a Caesar cipher.

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## 27. Email Security (PGP and S/MIME)

**Ans:**

**PGP** combines symmetric and public key encryption for secure email and digital signatures.

**S/MIME** uses **digital certificates** to provide encryption and authentication for company emails.

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## 28. Phishing

**Ans:**

Phishing is a **fake email or message** that looks real and tricks users into giving passwords or banking details.

Example – a mail saying “Update your bank info now”.

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## 29. Types of Authentication

**Ans:**

1. **Password-based:** Uses username and password.
  2. **Biometric-based:** Uses fingerprints or face recognition.
  3. **Card-based:** Uses smart or access cards.
  4. **Two-Factor:** Combines two methods like password + OTP for extra security.
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